Market Power and Contagion in Interbank Networks

An Agent-Based Model of Systemic Risk

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Outline

Abstract

- **Agent-Based Models (ABM)** to study systemic risk in interbank credit networks.
- Relationship between **network connectivity**, **market power**, and **bankruptcy cascades**.
- Key finding: a **non-monotonic trade-off** between risk sharing and systemic risk—connectivity diversifies risk in normal times but amplifies contagion during crises.
- 2008 financial crisis as an example of risk-sharing mechanism under high leverage and heterogeneous agents.

Outline

The 2008 Crisis: Background

- Increased bank leverage and corporate debt-equity ratios made the economy fragile.
- Traditional theory fails to explain complex agent interactions during crises.
- The interbank market became a **contagion vector**: failure of one bank triggers a domino effect through interlocking exposures.

The Risk-Sharing Trade-off

Two opposing effects of network connectivity (Tedeschi et al., 2012):

- **Benefit:** Increasing connectivity allows **risk sharing**—banks diversify exposure across more counterparties.
- **Cost:** Excessive connectivity increases **systemic risk**—default propagates through the network.
- Evidence: a **non-monotonic relationship** exists—low connectivity stabilizes by diluting shocks, but high connectivity facilitates cascading failures.

"Spreading the risk around the globe may improve stability in good times. However, in times of crisis, critical perturbations can spread across the whole system." – Tedeschi et al.

Empirical Evidence

- Allen & Gale (2000), Thurner et al. (2003), Iori et al. (2006): modeling credit systems as random graphs shows that bankruptcy avalanche probability **decreases** with connectivity.
- **Exception:** Lorenz & Battiston (2008) show that with **financial acceleration** (trend reinforcement in fragility), the probability of default does **not** decrease monotonically with connectivity.
- Iori et al. (2006): the root of avalanches lies in **agent heterogeneity**.
- Financial crises are characterized by the **procyclicality of leverage** across institutions.
- Reserve requirements dampen cascades but **limit growth** by reducing firm investment.

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Model Overview

- ABM with N banks in a sequential economy $t = \{0, 1, \ldots, T\}$.
- Banks connected through an **Erdős–Rényi** random graph $G(N, p)$: each edge exists with probability p.
- Three interacting sectors: goods market, credit market, interbank market.
- Goal: analyze how **connectivity** and **market power** affect system stability.

Balance Sheet

Accounting Identity

$$L_i^t + C_i^t + R_i^t = D_i^t + E_i^t$$

- **Assets:** Long-term loans (L), liquidity (C), reserves (R).
- **Liabilities:** Deposits (D), equity (E).
- Reserves set by regulation: $R_i^t = \hat{r} D_i^t$ with $\hat{r} = 0.02$.

Liquidity Shocks

Deposits are exogenously shocked each period:

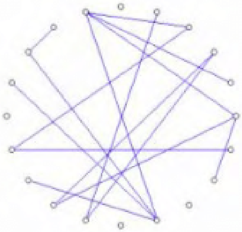
$$D_i^t = D_i^{t-1}[\mu + \omega \cdot U(0,1)]$$

- μ: expected growth rate; ω: shock magnitude.
- **Potential lenders:** banks with positive net deposits (after reserves).
- **Potential borrowers:** banks with negative net deposits exceeding liquidity.

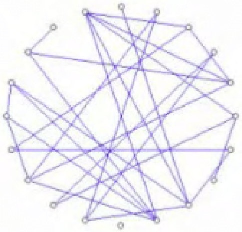
Network Structure

- Bilateral credit lines formed at the start of each period.
- Connectivity controlled by parameter p in the Erdős–Rényi graph.
- Each borrower is matched to **exactly one** lender per period.
- The **Giant Component Size (GCS)** measures network integration.

$p = 0$



$p = 0.1$



$p = 0.2$

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Expected Profit

Banks are risk-neutral in perfect competition. Expected profit from lending to borrower j:

$$E[\Pi_{i,j}^t] = p_j^t(r_{i,j}^t - m)c_{i,j}^t - (1 - p_j^t)m \cdot c_{i,j}^t$$

- m: screening cost (0.015).
- p_j^t: probability of borrower survival.
- $r_{i,j}^t$: interest rate; $c_{i,j}^t$: lending capacity.

Interest Rate Derivation

Setting $E[\Pi] = 0$ (zero-profit condition):

$$r_{i,j}^t = \frac{m}{p_j^t}$$

Introducing **market power** $\psi \in [0, 1)$:

$$\boxed{r_{i,j}^t = \frac{m}{p_j^t(1-\psi)}}$$

- $\psi = 0$: competitive (actuarially fair) rate.
- $\psi > 0$: monopolistic markup from information asymmetry.

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Endogenous Market Power

Market power evolves endogenously with bank size:

$$\psi_i^t = \frac{E_i^t}{E_i^{t,\max}}$$

- Larger banks (higher equity) exercise more market power.
- Borrower default probability: $p_j^t = E_j^t / E_j^{t,\max}$.
- Leverage: $\lambda_j^t = d_j^t / E_j^t$.
- The model evolves **only** from screening costs and shock parameters.

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Equity Evolution

$$E_t^i = E_{t-1}^i + \sum_j l_t^{i,j} r_t^{i,j} - \sum_{j \subseteq \theta_t^i} \left[B_t^{i,j} - \tilde{L}_t^j \right]$$

- **Bankruptcy:** bank exits if $E_i^t < 0$.
- **Fire sales:** last-resort option (blocked in this model to amplify contagion effects).
- Failed banks replaced by new entrants with small initial capital.

Failure Channels

Three distinct causes of bank failure:

1. **Rationing:** borrower cannot secure sufficient funding.
2. **Contagion:** lender suffers losses from borrower default (bad debt).
3. **Repayment failure:** fully-funded borrower cannot repay loan or interest.

Key finding: **repayment failure dominates** at high connectivity, while rationing dominates at low connectivity. Contagion remains minimal ($< 5\%$).

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Propagation Mechanisms

Three primary contagion channels:

1. **Bank runs:** self-fulfilling panics.
2. **Asset price contagion:** fire-sale spirals.
3. **Inter-locking exposures:** direct interbank linkages.

This model focuses on the **interbank market** as the primary liquidity crisis transmission mechanism.

The Risk-Sharing Trade-off

- **Benefit:** connectivity enables risk sharing and diversification.
- **Cost:** excessive connectivity amplifies systemic risk.
- Evidence: **non-monotonic relationship** between connectivity and systemic risk (with agent heterogeneity).

The literature shows that the systemic risk from connectivity can **outweigh** the insurance benefits of risk sharing.

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Phase Transition

The system exhibits a critical transition near $p \approx 0.05$:

- **Percolation threshold:** GCS jumps from ~ 1 to ~ 46 banks (92%).
- **Low p:** rationing-dominated, high interest rates, few loans.
- **High p:** repayment-failure-dominated, low rates, saturated network.

For $p > 0.05$, all macroeconomic metrics stabilize—the network is saturated.

Connectivity and Systemic Risk

- Increasing p: interest rate drops from 30 (cap) to ~ 1.65.
- Bankruptcies: stable at ~ 7–7.5 (limited by model structure).
- Active loans: saturate at ~ 13 (26% of banks).
- Equity: jumps from 46.8 to ~ 55 then plateaus.
- Bad debt: grows rapidly then stabilizes at ~ 1.3.

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Key Findings

1. **Phase transition at** $p \approx 0.05$: the network saturates and all metrics stabilize.
2. **Failure composition shifts**: rationing dominates at low p; repayment failure dominates at high p.
3. **Contagion is minimal** ($< 5\%$ of failures)—the dominant risk channel is business failure, not network propagation.
4. **Market power** (ψ) is positively correlated with rates and decreases as connectivity increases.

Policy Implications

- **Heterogeneity** is a root cause of systemic instability; failure of large banks is catastrophic.
- Interbank lending should ideally be restricted among banks with **similar liquidity characteristics.**
- Reserve requirements dampen cascades but **limit growth.**
- Risk diversification across many agents has **ambiguous** effects on total systemic stability.

Thank You